Hari Krishnan A J

Software Engineer

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SUMMARY

Backend and AI Engineer with 3+ years of experience shipping production RAG and LLM applications for B2B sustainability SaaS serving 15+ enterprise clients. Built a hybrid retrieval system reaching ~90% accuracy across 20M emission factors. Took an EU CBAM compliance solution from concept to market, and earlier managed 15 people at an AI startup.

EXPERIENCE

Karbunwise | Business Sustainability Intelligence Platform , Software Engineer March 2024 – Present | Hyderabad

LCA Platform

- Architected a four-stage hybrid retrieval pipeline over 20M emission factors, combining Postgres trigram and full-text search, OpenAI text-embedding-3 vectors in pgvector and FAISS, score fusion, and Voyage AI rerank-2.5 cross-encoder reranking
- Benchmarked matching accuracy at ~90% against completed customer projects using prior expert mappings as ground truth, backed by ongoing random-sample QA with human checks
- Devised a feedback loop that learns from corrected and historical matches to personalize ranking per account, reducing hand-mapping effort ~60% on a typical 500-input assessment
- Delivered a RAG-powered analytics dashboard replacing static reports with natural-language queries, on-demand charts and technical insights for every customer
- Led 6 engineers through a 2-month rebuild, redesigning backend services, database schema and API layer to resolve performance bottlenecks
- Migrated 10 existing accounts to the new multi-tenant system with zero downtime, supporting growth to 15
- Owned architecture decisions for the rebuild while assigning work, reviewing code and contributing to the core codebase
- Implemented an ingestion pipeline consolidating 50 sources into a single reference library, with new feeds added in hours rather than days
- Automated PCR ingestion, cutting per-document setup time ~70% and expanding coverage to 100+ PCRs
- Co-developed an Autodesk Revit plugin that brings building project details directly into the application, shrinking imports from hours to minutes
- Launched a self-service trial portal that has signed up 200+ users across several regions

EU CBAM Compliance

- Headed a 3-person team that produced the CBAM solution from scratch in 3 months, spanning supplier data collection, carbon calculations and submission-ready EU reports for all 6 sectors
- Translated EU regulations into product requirements, data models and endpoints, compressing per-client preparation from weeks to days
- Streamlined supplier emissions intake with a precursor-identification mechanism that applies default values, eliminating ~70% of manual entry
- Onboarded 5 exporters across India and the Middle East, refining workflows with sustainability consultants before handover

Interain AI | Recruitment Automation Platform, Lead Software Developer August 2023 – February 2024 | Chennai

- Joined as first technical hire and guided a 15-member team, including 6 developers, to release an AI interview-preparation tool on Next.js and FastAPI
- Designed an adaptive AI engine that generates difficulty-calibrated questions from each student's history, pinpoints strengths and weaknesses, and returns targeted feedback to 100 active users
- Created practice sessions, tests and progress tracking so students could monitor performance over time
- Secured partnerships with 5 Chennai colleges through live demos and trials, converting pilots into early adopters

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++
AI & GenAI: Retrieval-Augmented Generation (RAG), hybrid search, OpenAI text-embedding-3, pgvector, FAISS, Voyage AI cross-encoder reranking, retrieval evaluation, prompt engineering, LLM integration
Backend: FastAPI, Flask, Django, REST API design, ETL pipelines
Databases: PostgreSQL (full-text search, pg_trgm, query optimization, schema design), Redis, MongoDB, MySQL
Cloud & DevOps: AWS (EC2, S3, RDS), Docker, Nginx, Linux, Jenkins, CI/CD, Flyway, Git

EDUCATION

Shiv Nadar University, B.Tech Computer Science and Engineering with Specialisation in IoT 2021 – 2025 | Chennai
CGPA : 8.8/10